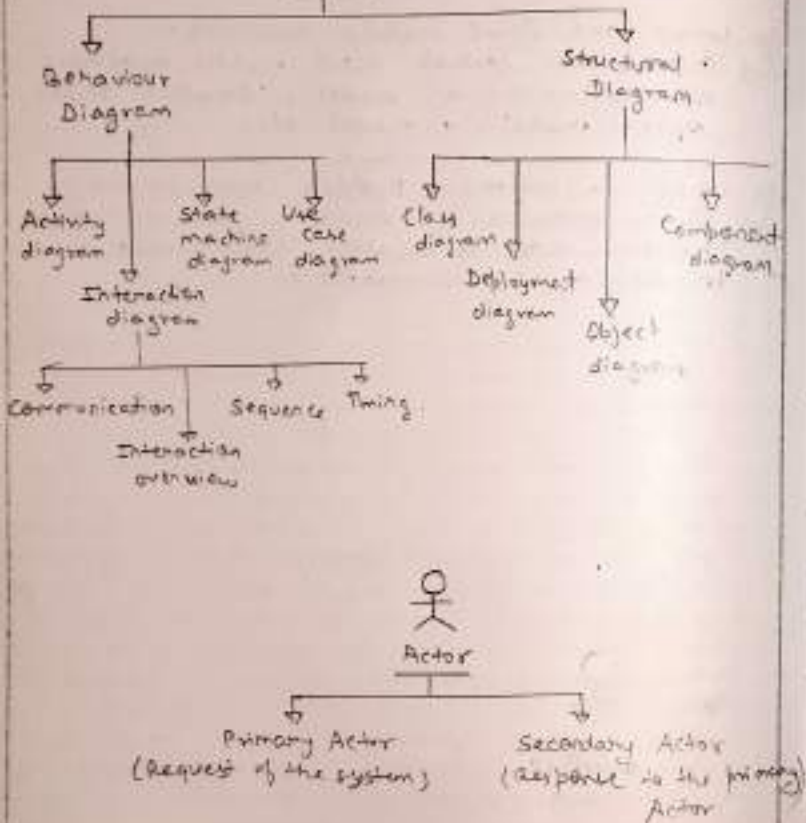


UML



Experiment-3

Aim To understand the concept of UML (Unified modeling language) & draw Use Case diagrams with their relationships for software development.

Theory is

UML - Unified modeling language is a standard modeling language used to visualise, specify, construct & document the structure & behaviour of a software system.

It provides both static (structural) & dynamic (behaviour) views of the system.

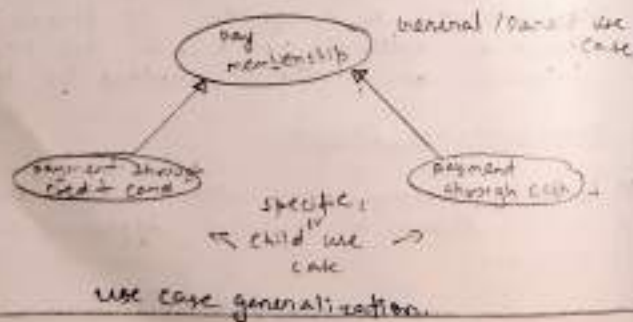
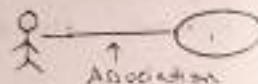
UML is widely used in software development for designing & analyzing system before actual implementation.

* Use case Diagram:- A use case diagram is used to represent the functional requirements of a system. It shows actors (Users or external system) & use cases (services or functional provided by the system).

Representation Elements

(1) Actor:- External entity interaction with the system, shown as a stick figure.

Use Case



12) Use Case - A system function, represented by an oval. Perform some task for Actor.

13) Association - A solid line between actor to use case.

14) Include - Shows mandatory reuse of another use case. (to avoid repetition)
Notation => <<include>>

Example "Login" use case may include "validate user".

15) Extend - Represent optional / conditional behaviour
Notation => <<extend>>

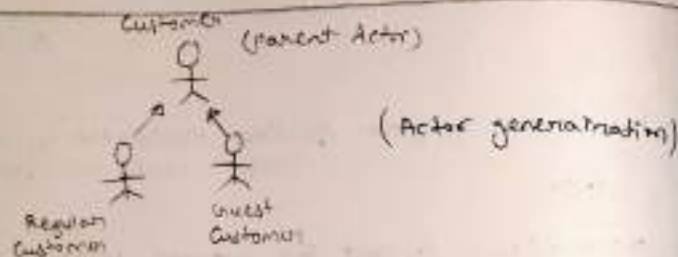
Example - "Place order" may extend "Apply discount".

16) Generalization - Shows inheritance between actors or use cases.

- Represented by a solid line with a hollow arrowhead pointing to the parent.

- Example -

"Registered user" & "Guest user" can both generalize the actor "customer".



LMS Library management system



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→ It gives how a child can specialize behaviour characteristics from parent.

Process

- (1) Identify actors (User, systems interacting with software)
- (2) Identify use cases (functional requirements)
- (3) Draw system boundary (Rectangle)
- (4) Place actor outside boundary & connect them to the use cases.
- (5) Add relationships (association, include, extend, generalization) where required.

Result:- The concept of UML & use case diagrams were studied. The diagram was successfully drawn showing actors, use cases & relationships (association, include, extend, generalizations).

Teacher's Signature _____